

PS7_Zavala

Emilian Zavala

March 31st 2026

Step 6

Below is a summary table of the wages data frame. Approximately 25% of `logwage` observations are missing. This is most likely MNAR (Missing Not At Random), as women with lower wages may be less likely to report them.

Table 1:

Statistic	N	Mean	St. Dev.	Min	Max
logwage	1,669	1.625	0.386	0.005	2.261
hgc	2,229	13.101	2.524	0	18
tenure	2,229	5.971	5.507	0.000	25.917
age	2,229	39.152	3.062	34	46

Step 7

Below are the results of four imputation methods. The true value of $\hat{\beta}_1 = 0.093$.

Step 8

For my project, I am using a baseball dataset containing statistics from the modern era (post-1900). The dataset includes offensive and pitching statistics such as batting average, home runs, RBI, ERA, strikeouts, and WAR (Wins Above Replacement). My primary modeling approach will be a linear regression model predicting WAR from key offensive and pitching statistics, allowing me to identify which performance metrics are most strongly associated with overall player value. I may also explore separate models for hitters and pitchers, as WAR is calculated differently for each.

Table 2:

	<i>Dependent variable:</i>		
	logwage	logwage_mean	logwage_pred
	(1)	(2)	(3)
hgc	0.062*** (0.005)	0.050*** (0.004)	0.062*** (0.004)
collegenot college grad	0.145*** (0.034)	0.168*** (0.026)	0.145*** (0.025)
tenure	0.050*** (0.005)	0.038*** (0.004)	0.050*** (0.004)
I(tenure ²)	−0.002*** (0.0003)	−0.001*** (0.0002)	−0.002*** (0.0002)
age	0.0004 (0.003)	0.0002 (0.002)	0.0004 (0.002)
marriedsingle	−0.022 (0.018)	−0.027** (0.014)	−0.022* (0.013)
Constant	0.534*** (0.146)	0.708*** (0.116)	0.534*** (0.112)
Observations	1,669	2,229	2,229
R ²	0.208	0.147	0.277
Adjusted R ²	0.206	0.145	0.275
Residual Std. Error	0.344 (df = 1662)	0.308 (df = 2222)	0.297 (df = 2222)
F Statistic	72.917*** (df = 6; 1662)	63.973*** (df = 6; 2222)	141.686*** (df = 6; 2222)

Note:

*p<0.1; **p<0.05; ***p<0.01